Angchen Xie

Homepage | xieangchen808@gmail.com | Google Scholar | GitHub | X | Pittsburgh, PA

Education

Carnegie Mellon University — Pittsburgh, PA
M.S. in Robotics (MSR) — *Aug. 2025 – May 2027 (expected)*
Advised by Guanya Shi and Max Simchowitz

- GPA: 4.17/4.00
- Courses: Introduction to Robot Learning; Optimal Control and Reinforcement Learning; Computer Vision; Advanced Control for Robotics; Mechanics of Manipulation

Shanghai Jiao Tong University — Shanghai, China
B.Eng. in Automation — *Sept. 2021 – June 2025*

- GPA: 4.01/4.30 | **Rank: 1/87**
- Undergraduate thesis: *Fault-Tolerant Control of Legged Robots under Multi-Sensor Fusion*

Research Interests

I work toward general-purpose robots that combine broad learned priors with grounded models of themselves and the physical world, enabling reliable and versatile behavior across tasks, embodiments, and environments.

Publications

* Equal contribution.

[1] **[Under review, CoRL 2026] Angchen Xie***, Nikhil Sobanbabu*, Ishayu Shikhare, Alan Wang, Max Simchowitz, Guanya Shi. *“FADA: Few-Shot Domain Adaptation via Dynamics Alignment for Humanoid Control”*.
arXiv Website Thread

[2] **[RSS 2026]** Rishabh Madan, **Angchen Xie**, Samantha Saak, Andres Blanco, Dohyeok Lee, Sarah Grace Brown, Yunting Yan, Mark Zolotas, Jose Barreiros, Tapomayukh Bhattacharjee. *“TACTIC: Tactile and Vision Conditioned Contact-Centric Control for Whole-Arm Manipulation”*.
arXiv Website

[3] **[RSS 2026]** Pablo Ortega-Kral*, Eliot Xing*, Arthur Bucker, Vernon Luk, Junseo Kim, Owen Kwon, **Angchen Xie**, Nikhil Sobanbabu, Yifu Yuan, Megan Lee, Deepam Ameria, Bhaswanth Ayapilla, Jaycie Bussell, Guanya Shi, Jonathan Francis, Jean Oh. *“RIO: Flexible Real-Time Robot I/O for Cross-Embodiment Robot Learning”*.
arXiv Website Code

[4] **[CoRL 2025]** Rishabh Madan, Jiawei Lin, Mahika Goel, **Angchen Xie**, Xiaoyu Liang, Marcus Lee, Justin Guo, Pranav N. Thakkar, Rohan Banerjee, Jose Barreiros, Kate Tsui, Tom Silver, Tapomayukh Bhattacharjee. *“PrioriTouch: Adapting to User Contact Preferences for Whole-Arm Physical Human-Robot Interaction”*.
arXiv Website

[5] **[IROS 2024] Angchen Xie**, Yeqiang Qian, Weihao Yan, Chunxiang Wang, Ming Yang. *“Non-repetitive: A Promising LiDAR Scanning Pattern”*.
Paper Dataset

Research Experience

LeCAR Lab, Carnegie Mellon University — Pittsburgh, PA
Research Intern — *Aug. 2025 – Present*
Advised by Guanya Shi and Max Simchowitz
Topic: humanoid whole-body control; infrastructure for cross-embodiment robot learning

- A Planner-IDM (inverse dynamics model) architecture that adapts to unseen terrain, payload, and actuator dynamics by finetuning only the IDM on about two minutes of target-domain rollouts, without rewards or demonstrations. *(FADA)*
- Open-source infrastructure validated on vision-language-action deployment across three morphologies and four hardware platforms. *(RIO)*

RL^2 Lab, Shanghai Jiao Tong University — Shanghai, China
Research Intern — *Jan. 2025 – June 2025*
Advised by Yue Gao
Topic: fault-tolerant locomotion for legged robots

- Model-based reinforcement learning for hexapod locomotion across varied terrain, deployed on hardware.
- A controller that adapts to the failure of one or more joints, including failure modes unseen during training.
- Multi-sensor fusion for terrain and self-failure perception, improving generalization on the real platform. *(Undergraduate thesis)*

EmPRISE Lab, Cornell University — Ithaca, NY
Research Intern — *July 2024 – Dec. 2024*
Advised by Tapomayukh Bhattacharjee
Topic: contact-rich whole-arm manipulation for physical human–robot interaction

- A contact-centric predictive model coupling learned latent dynamics with analytical kinematics through contact Jacobians, driven by a contact-aware sampling-based MPC planner. *(TACTIC)*
- A learning-to-rank framework that resolves conflicting user force preferences across simultaneous contacts. *(PrioriTouch)*

CyberC3 Intelligent Vehicle Lab, Shanghai Jiao Tong University — Shanghai, China
Research Intern — *Nov. 2022 – June 2024*
Advised by Yeqiang Qian and Ming Yang
Topic: LiDAR perception for intelligent vehicles

- *Repetitive-or-not,* a CARLA dataset captured simultaneously by repetitive and non-repetitive LiDARs, isolating scanning pattern as the single variable.
- Showed non-repetitive scanning is the more promising pattern on the strength of its object scanning, raising the maximum detection count on small traffic participants by 59.6%, and quantified the domain gap between the two point-cloud types. *(Non-repetitive)*

Service

Reviewer, Conference on Robot Learning (CoRL)	2026
Reviewer, Robotics: Science and Systems (RSS)	2026
Reviewer, IEEE Robotics and Automation Letters (RA-L)	2025, 2026

Honors and Awards

Outstanding Graduate Award, Shanghai Jiao Tong University	2025
China National Scholarship (top 0.2%)	2024
First Prize Academic Scholarship, Shanghai Jiao Tong University	2024
National First Prize, China Undergraduate Electronics Design Contest (top 1%)	2023
First Prize, China Undergraduate Mathematical Contest in Modeling (top 2%)	2022